

Assignment

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Subject: SEPM

Q To understand DevOps: Principles, practices & DevOps Engineer Roles & Responsibilities

DevOps is a Combination of Development (Dev) & Operations (Ops). It is a culture, set of practices and tools that improves Collaboration between software developers & IT operations team. The main goal of DevOps is to deliver software faster, with better quality and with continuous improvement.

* Principles of DevOps

1. Collaboration and Communication.

Development, testing and operations teams work together throughout the software lifecycle.

2. Automation

Repetitive tasks like testing, building, deployment and monitoring are automated to save time & reduce errors.

3. Continuous Integration (CI)

Developers frequently merge code into a shared repository where builds and tests run automatically.

4. Continuous Delivery / Continuous Deployment (CD)

Software updates are released quickly and reliably through automated pipelines.

5. Monitoring and Feedback
Applications and infrastructure are continuously monitored to detect issues and improve performance.

6. Customer centric approach
Focus is on delivering value to users faster with frequent updates and improvements.

7. Infrastructure as Code (Iac)
Servers, networks and environment are managed using code instead of manual configuration.

* Dev Ops Practices.

- Continuous Integration (CI)
- Continuous Delivery (CD)
- Continuous Deployment
- Version Control
- Automated testing
- Configuration management
- Containerization
- Orchestration
- Monitoring and logging

* Roles of a DevOps Engineer.

A DevOps engineer is responsible for bringing the gap between development and operations teams. They automate workflows, manage infrastructure and ensure reliable software delivery.

- * Responsibility of a DevOps Engineer.
 - Builds and Release management. Create automated build and development pipelines.
 - CI/CD pipeline setup: use tools like Jenkins, Github, actions or Github CI/CD.
 - Infrastructure management: provision and maintain servers, cloud resources and environment.
 - Automation: automate testing, deployments, backups and repetitive tasks.
 - Monitoring and Incident Response.
 - Security Integration (Dev Secops): Include security checks in pipelines and infrastructure.
 - Collaboration: coordinate between developers, QA, and operations teams.
 - Performance Optimization: improve scalability, reliability, etc.
 - Documentation: maintain runbooks, deployment guides and architecture documents.